

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester V Examination, 2022-23

Botany**Paper: BOB-503 — Plant Ecology, Environmental Pollution and Toxicology (2016)**

Time: Three Hours

Maximum Marks: 70

Instructions: Answer five questions selecting two each from Sections A and B and Question 1 which is compulsory. The figures in the right-hand margin indicate marks.

Question 1. Answer/describe/explain briefly, in about 50 words each of the following:
[10 × 1 = 10]

- (a) What is a lux meter?
- (b) What do you understand by DBH?
- (c) What is the relationship between altitude and atmospheric temperature?
- (d) Differentiate between keystone and dominant species.
- (e) Define community stability.
- (f) What are criteria air pollutants?
- (g) State the foliar injury symptoms of SO₂.
- (h) Differentiate between effective dose and lethal dose.
- (i) Give an example of a Cd hyper-accumulator plant.
- (j) Define NOEL and LOEL.

SECTION A

Question 2. Define productivity. Briefly discuss the global patterns of primary productivity and its relation to litter fall. [15]

Question 3. Write a comprehensive account of any two of the following: [2 × 7.5 = 15]

- (a) Saprophagic-Biophagic energy flow model
- (b) MacMahon's model of ecological succession
- (c) Continuum concept of ecological succession

Question 4. Write critical notes on any two of the following: [2 × 7.5 = 15]

- (a) Mechanisms of population differentiation

- (b) Species diversity
- (c) Ecological niche

Question 5. Write a detailed account of the population dynamics and its regulation. [15]

SECTION B

Question 6. Give any two important effects of air pollutants on the plants. Discuss the formation of different secondary air pollutants in the atmosphere. [15]

Question 7. Write short notes on any two of the following: [2 × 7.5 = 15]

- (a) Biomagnification and bioaccumulation
- (b) Thermal water pollution
- (c) Eutrophication

Question 8. What is the contribution of Rachel Carson? Describe briefly the environmental fate of pesticides and their impacts on ecosystems with the help of suitable examples. [15]

Question 9. Write short notes on any two of the following: [2 × 7.5 = 15]

- (a) Types of toxicants and their ecological implications
- (b) Dose response relationships
- (c) Factors affecting toxicity

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